

Advanced Digital Design Lab
130326
Traffic Light Controller (Alternative Assessment)
Date: Tuesday, June 10, 2025 (Modified: June 20, 2025)

Submission:	By not more than 2 people together
Deadline:	As instructed by the lecturer– via moodle
Note:	שימו לב: (1) משימה זו מיועדת להערכה חלופית ומחליפה את המבחן הסופי של הקורס (30% מהציון הסופי) (2) לפי התקנון, יש להשיג במרכיב מסוג זה ציון של לפחות 55% (3) מטלה זו תוצג בפני צוות הקורס תוך תיאום מראש (4) במעמד ההערכה החלופית והצגת התוצאות, צוות הקורס ישאל גם שאלות ידע והבנה כלליות בנושאי הקורס (5) יש להעלות את מסמך הביצוע של המטלה (דו"ח) לתיבת ההגשה According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment

This lab contains VHDL simulation and synthesis of VHDL state-machine for traffic lights.

The following skills are exercised in the course of this lab:

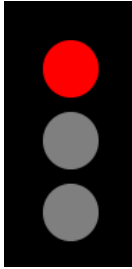
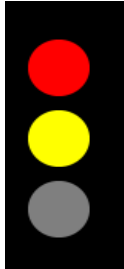
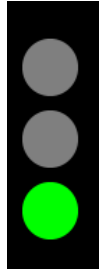
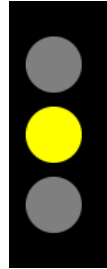
- Finite State Machines (FSM) – See VHDL lecture 8 (“State Machines”)
- Hierarchical Design – See VHDL lecture 9 (“Hierarchy”)
- Skills of a designer – Earned by careful experience

Read the lab guidelines completely and carefully before doing any design work. Do not write any VHDL code prior to drawing complete and clear block diagrams and state diagrams

A. Design Specifications

- One lane, one way, motorway "traffic light" for CARS (colors of RED, YELLOW, and GREEN);
- PEDESTRIANS' "traffic light" (RED, GREEN);

Transitions (See [1], [2]):

	<i>Step 1</i>	<i>Step 2</i>	<i>Step 3</i>	<i>Step 4</i>
Cars	 Red	 Red + Yellow	 Green	 Yellow
Pedestrians	 Green	 Red	 Red	 Red
Duration	10 seconds	2 seconds	10 seconds	2 seconds

After reset, the traffic lights cycles **start from “Step 1”** and repeat endlessly.

Electrical/Logical requirements:

- Clock: **100 MHz**
- Active high synchronous reset
- Design is fully synchronous
- Output: 5 bits, driving 5 LEDs that represent the appropriate traffic lights

The design is targeted to the Basys3 board:

- The only inputs to the design are the **100 MHz** clock and the Reset – on board push-buttons shall be used
- The only outputs of the design are the 5 signals that drive the LEDs – on board LEDs shall be used

B. Design guidelines and preparation work – (Be aware of the due date !):

Objective:

Create a hierarchical design built from the following components:

1. Control unit based on a Moore state-machine component (The state machine is with a "clock enable" signal).
2. Hard coded Time-base generator for generating the "clock enable" signal every 1 second.
3. Two counters – one that counts 2 seconds and another one that counts 10 seconds (subject to clock and "clock enable" signals). You can have the two counters as two VHDL processes in a single module.

Note:

The hierarchical design contains a number of modules. Throughout this preparation work, we ask you to:

- Write an ATP for the module, as specified in the requirements.
- Create a VHDL testbench for the module, based on the module's ATP.

This is the standard flow to reduce risks when integrating the modules altogether into the final "top level" design. We recommend that you do so...

However, if you want to save time, and do the ATP and testbench for the "top level" module only (i.e., skipping ATP and testbench for sub-modules), you may do so, but... by this you are taking a risk of using unchecked modules that you haven't guaranteed their proper functional behavior. Moreover, should an error occur in a sub-module at a later stage you might need to do the ATP and testbench anyway while debugging the problem.

B1. VHDL entity

- Draw the design as a "black box" with I/O signals.
- Write a table describing the I/O signals
- Write a VHDL entity for the top level design
- Write appropriate XDC file for the design

B2. Draw a block diagram for the design.

Show input and internal signals clearly, indicating signal name and width appropriately.

B3. Control Unit (State Machine component):

Design a VHDL "Moore" state machine for the traffic light system. The state-machine must have a "clock enable signal".

- The state machine is responsible to drive (assign) the traffic light outputs
- The state machine is responsible to assign the start counter signals for each counter
- The state machine receives the TC signals from the counters as inputs

- Define the control unit's entity
- Draw (manually) the bubble diagram
- Write VHDL code for the machine
- Write an ATP for the state machine design, as specified in the requirements.

- Create a VHDL testbench for the control unit, based on the above ATP.
- Run VHDL simulation of the control unit design.

B4. Time-base generator component:

The time-base component creates a "clock enable" signal for the state-machine once every 1 second after reset, based on the **100 Hz** input clock. You are encouraged to reuse previous hard-coded time-base generator design that you wrote in any previous lab assignments.

- Write an ATP for the Time-base generator components
- Write a dedicated testbench to the Time-base generator according to the ATP (You are encouraged to reuse testbench written in any previous lab assignments, but make sure that it is according to the ATP).
- Simulate the time-base generator using the testbench that you created for it.

B5. Counters component(s)

The counter(s) component contain two up-counters:

- One counter counts 2 (i.e., 0, 1), has a start counter signal and a TC signal
- The other counter counts 10 (i.e., 0, 1, 2.....9), has a start counter signal and a TC signal

B6. Top-Level design simulation

- Write an ATP for the top level design
- Create a top-level testbench based on the ATP. Make any possible effort to reuse one of the above testbenches

<end of preparation work...>

C. Simulation

- Run source simulation of 2 or 3 traffic light cycles.

D. Synthesis:

- Target device: The Artix-7 device of the Basys3 board
- Target Frequency: **100 MHz**
- Be aware: this is hierarchical design.

E. P&R and Post P&R Simulation:

- Place & Route the design.
- Perform post P&R gate-level simulations similar to the source simulation

F. Your lab report must include:

Note:

Some of the following points (e.g., block diagrams, ATPs, testbenches etc) were already required and submitted in the course of the preparation work. Nevertheless, given possible changes that occurred during the lab work, we are asking you to submit the final version of these in the course of your final report for this lab.

- All block diagrams and state diagrams that were required in section B of this lab. **You will probably need to scan manual drawings into the report – this is OK.**
- All ATP specifications, both for the component levels and for the top level, as required in the preparation work (**see relevant note for sub-modules' ATP**).
- VHDL source code and testbenches both for the component levels and for the top level (**see relevant note for sub-modules' testbench**).
- ModelSim waveforms from source simulation, showing a full “traffic light sequence” from “RED” to “RED”.
- Synthesis timing and are report
- Synthesis log that reports about the state-machine encoding
- Place & Route area, timing and pad reports
- ModelSim waveforms from **post P&R simulation** (Gate level with SDF), showing a full “traffic light sequence” from “RED” to “RED”.



Illustration 1: Crossing the road according to the traffic lights design [3]

G. Board Level Implementation

Download the design into the Basys3 board. Create a short video clip that documents your implementation and include the link to the video clip in your lab report (as in a previous board implementation lab).

Pictures taken from:

[1]- http://en.wikipedia.org/wiki/Traffic_light

[2]- <http://sv.wikipedia.org/wiki/Trafiksignal>

[3]- https://en.wikipedia.org/wiki/Abbey_Road